Posterior sampling of NZSHM22 rupture rates

Workflow summary for the 211,000-rupture run with the band (very conservative preset)

Run folder: `NZSHM22_Matrices-211k/SLIP_SECTIONSauto_MFD_UNCERTAINTYas-given_DOWN_THE_MIDDLE_False_preset_very_conservative_floor-10_band10_1793438237`
Run started 31 October 2026, 22:17, and finished 1 November 2026, 07:41; posterior figures redrawn 2 November 2026 with each kept draw shown as an x.

Overview

The NZSHM22 inversion asks for the annual rate of every rupture in a rupture set such that the predicted slip rates, the regional magnitude-frequency distribution (MFD) and the paleoseismic event rates match their targets within their uncertainties: a linear system $G\,m \approx d$ with each row already divided by its uncertainty. The NZSHM22 solutions find one such model by simulated annealing. This workflow instead maps the whole set of models consistent with the data and a stated prior: a Bayesian posterior, sampled by an exact Gibbs sampler, from which it reports posterior means and spreads of every rupture rate, of every predicted datum and of sums such as fault event rates, together with weighted representative models for hazard calculations. The sections below follow the notebook `cru_workflow.ipynb` step by step; each figure is one this run wrote.

1. Inputs and configuration

The inputs are the NZSHM22 211k inversion matrices: `cru_A.txt` (G, 8,928 rows by 211,042 ruptures, 9.4 million nonzeros) and `cru_D.txt` (d). The rows are the subsections' slip rates, 18 MFD bins of 0.1 magnitude units from M6.8 to M8.6, 65 paleoseismic event rates, and penalty rows that force some ruptures to zero. The NZSHM22 solution for the same rupture set (`NZSHM22_InversionSolution-211k`) supplies the fault geometry for the figures, the named faults, and a reference model drawn in black on every figure. One configuration cell sets everything; this run used the defaults adopted in early November 2026:

setting	value	meaning
SLIP_SECTIONS	'auto'	each fault section's slip rate counts as one measurement, however many subsections it is copied to
MFD_UNCERTAINTY	'as-given'	the MFD bins keep the uncertainties the inputs give them (9.9% to 18.9%)
BAND	10	every prediction kept within a factor 10 of its target (a hard constraint)
DOWN_THE_MIDDLE	False	no balance term: the Gaussian misfit alone
FLOOR	−10	the lowest rate an active rupture can take, 10^{-10} per year
PRESET	'very_conservative'	3 independent replicates × 16 chains × 120,000 sweeps (burn-in 10,000, a draw kept every 32)
REPRESENTATIVES	100	weighted representative models selected for hazard

2. Cleaning and diagnostics

Cleaning (`preliminaries.py`) turns the raw system into the problem the sampler sees. The 16 penalty rows with entries of 10^{20} fix 6,203 small ruptures at zero, and 301 soft zero-forcing rows fix 301 more, so 6,504 ruptures are removed (all at rate 0 in every output). 6,216 rows that then hold no remaining rupture add a constant χ^2 of 80.4 that no model can change, and are dropped. What remains is 2,395 rows by 204,538 ruptures: 2,312 slip-rate rows, 18 MFD bins and 65 paleo rows. The slip-rate rows come in 507 runs of identical values, one per fault section; each section's rows are divided by √k for its k subsections, so a section's slip rate counts once rather than k times. The MFD bin targets run from 3.4×10^{-4} to 4.6×10^{-3} per year and sum to 0.0372 per year. If the stated uncertainties are right, a model drawn from the posterior should have χ^2 near 590 ± 34 (507

slip-rate measurements and 83 other rows): this is the *fit check* used below. The diagnostics (Figure 1) show how weakly individual rates are determined: G has rank 2,224, and the median diagonal of the resolution matrix is 0.005, so the data fix sums of rates far better than single rates. There are also 1.25 million pairs of near-duplicate ruptures (cosine similarity above 0.99), which the sampler handles with swap moves.

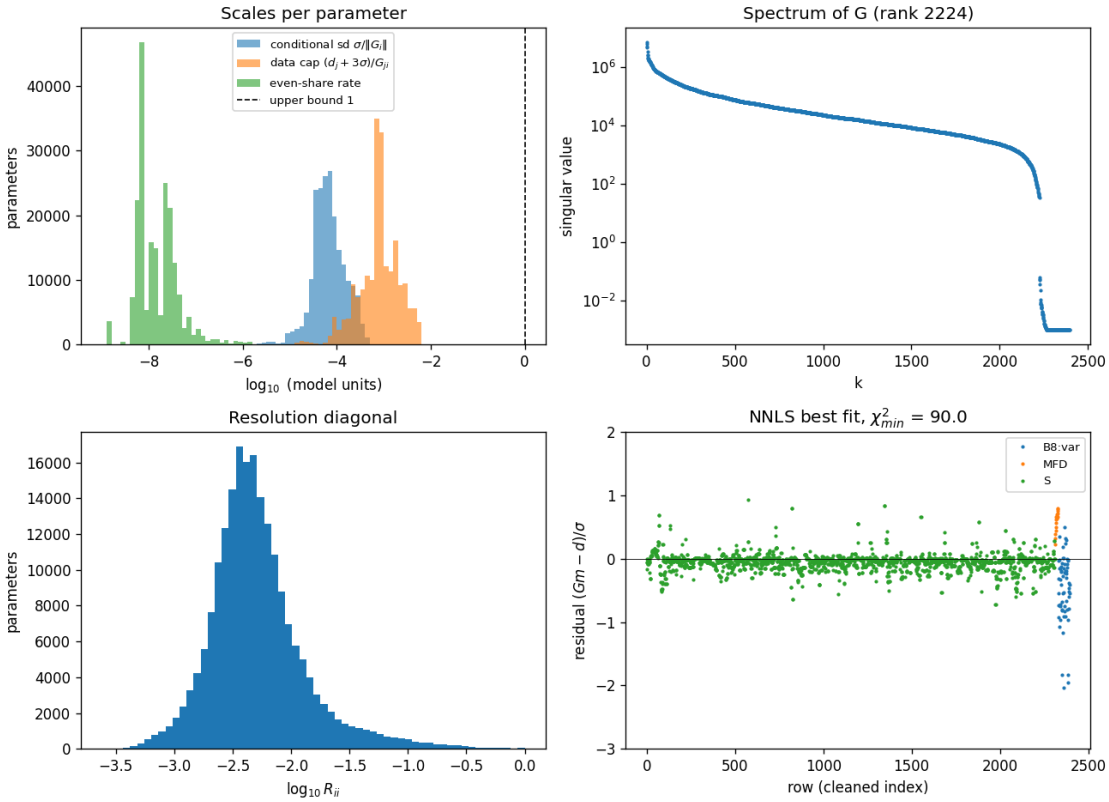


Figure 1. Cleaning diagnostics. Top left: per-rupture scales (the rate at which one rupture alone moves a prediction by one uncertainty, the most any datum allows it, and an even share of each datum). Top right: the singular values of G (rank 2,224). Bottom left: the resolution diagonal. Bottom right: the best fit's residuals, row by row, coloured by data type. `prelim.png`

3. The best fit (no prior)

Cleaning also computes the best nonnegative least-squares fit (Lawson–Hanson NNLS, 26 s). It reaches $\chi^2 = 90.0$ over the 2,395 rows with only 730 ruptures above zero: 50.7 from the slip rates, 6.8 from the MFD bins and 32.5 from the paleo rows. Its total rate is 0.0394 per year, 6.0% above the bin targets, with every bin between 2.3% and 13.9% above its target. Its χ^2 is far below the fit check because a fit chooses among 204,538 rates to match 2,395 numbers; it is a reference with no prior and the starting point of the chains, not a sample of the posterior. Figures 2–4 show its fit to each data type, with the NZSHM22 solution for comparison; the same three figures for the posterior follow in step 9. The best fit leaves 121 subsections outside the band (predicted at or near zero), so the chains cannot start from it as it stands (step 5).

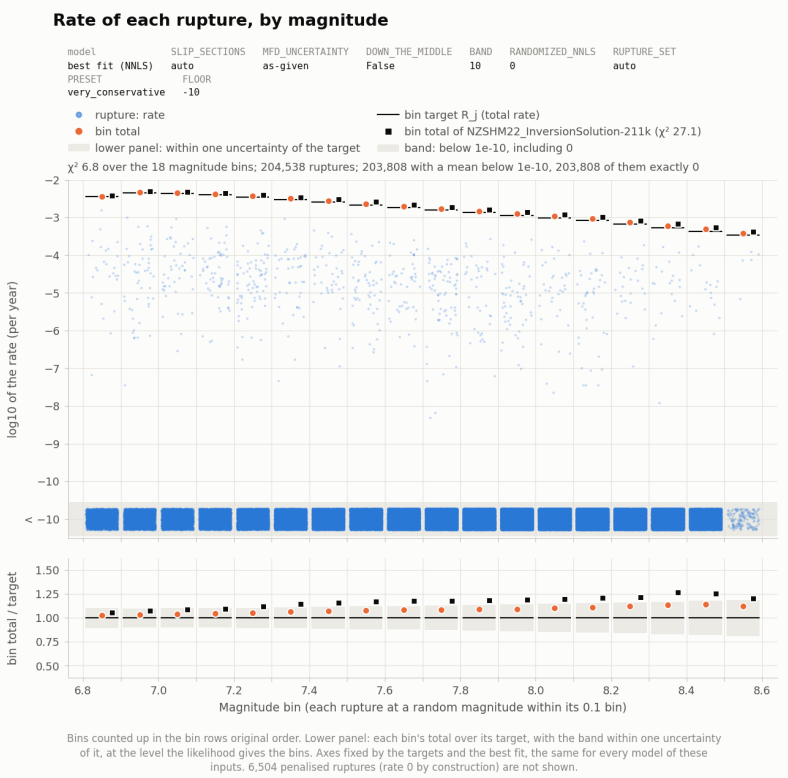


Figure 2. Best fit: each rupture's rate against its magnitude bin (the 203,808 ruptures at exactly zero in the band at the bottom), each bin's target (black line) and total (orange), and NZSHM22's bin totals (black squares). Lower panel: each bin's total over its target, with the band within one uncertainty. `nnls/fig_rate_magnitude_nnls.png`

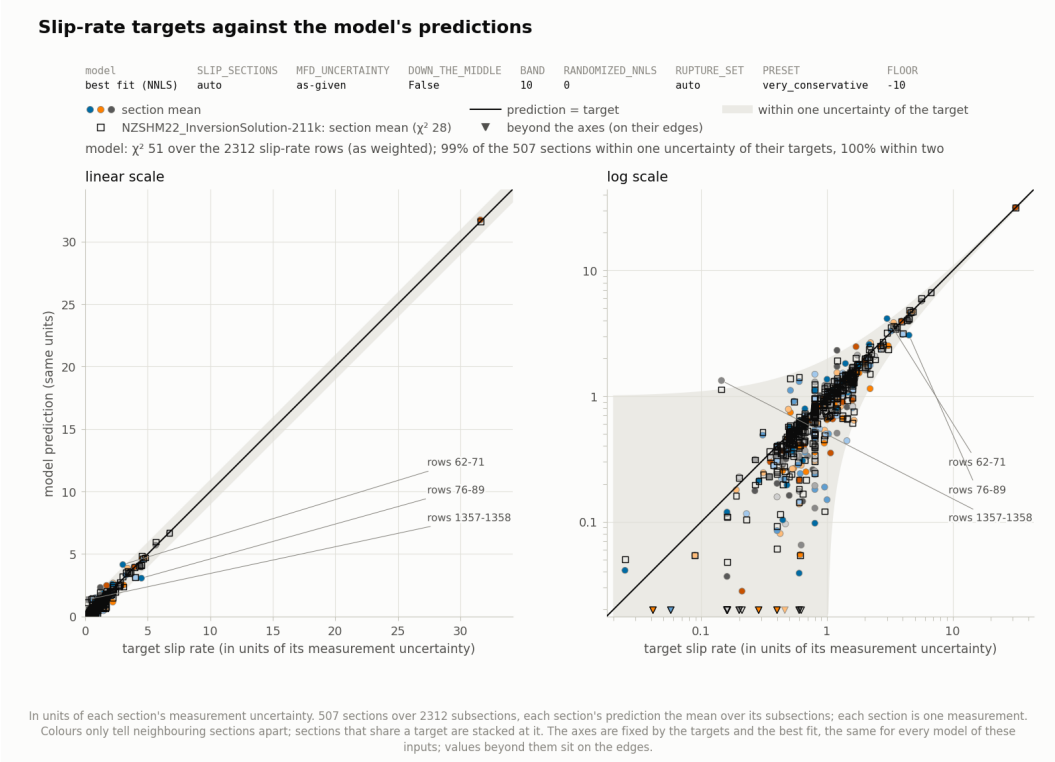


Figure 3. Best fit: each fault section's slip-rate target against the predicted section mean, in units of the section's measurement uncertainty, on linear and log scales; open squares are NZSHM22's predictions. The grey band is within one uncertainty of the target. `nnls/fig_slip_rates_nnls.png`

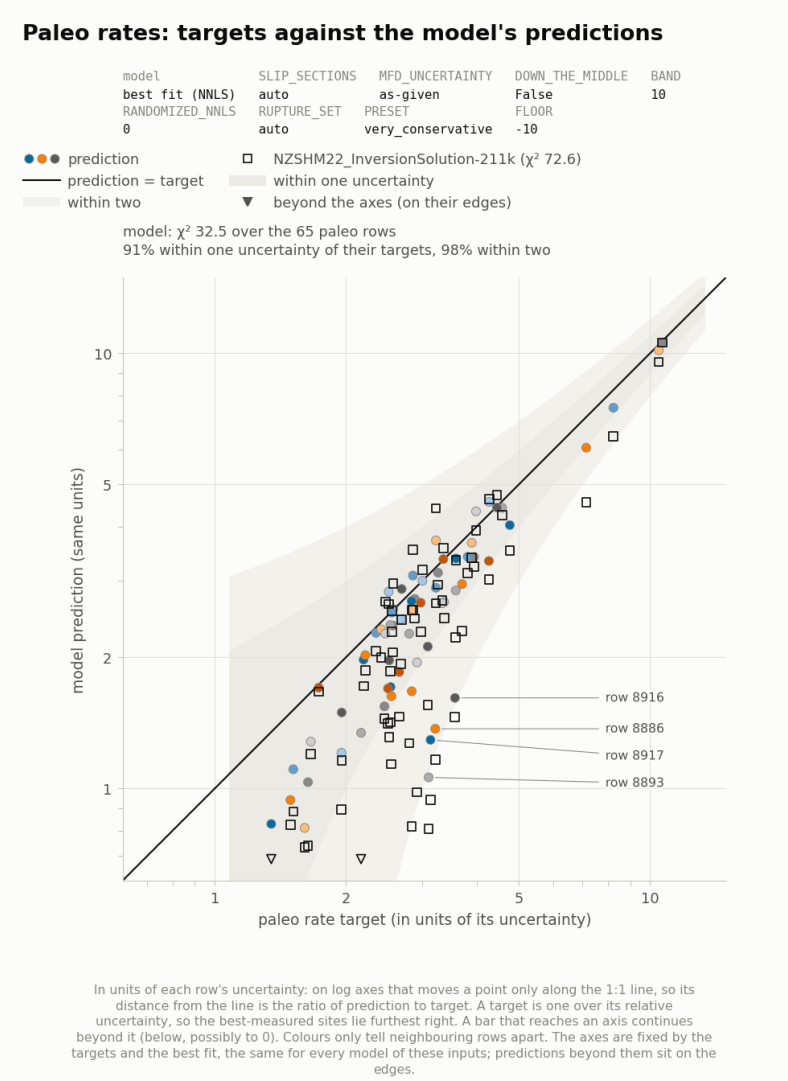


Figure 4. Best fit: each paleo row's target against its prediction, in units of its uncertainty, log-log; the shaded bands are within one and two uncertainties. `nnls/fig_paleo_rates_nnls.png`

4. The model: prior, likelihood and band

Prior. Each rupture is either off, with a rate of exactly zero, or on, with a rate that is uniform in log rate from 10^{-10} per year up to its magnitude bin's target rate (spike-and-slab). The top is soft, a Gaussian shoulder as wide as the bin's own uncertainty, since no single rupture can carry more than its bin's total. The fraction of ruptures switched on is estimated separately in each bin (uniform prior on each fraction), so the data decide how many ruptures are active, and the prior puts no pressure on the bin totals. **Likelihood.** Gaussian, $\exp(-\chi^2/2)$ with the stated uncertainties. **Band.** In addition, every prediction with a positive target (each subsection's slip rate, each MFD bin and each paleo rate) must lie within a factor 10 of its target, from $d/10$ to $10\,d$. This is a hard constraint, a prior that is zero outside it. Its main effect is on faults that share few ruptures with others: without it the posterior often leaves such faults with no events at all, since their weak slip-rate targets tolerate a prediction of zero; with it every subsection must slip at no less than a tenth of its target rate. The factor 10 is a modelling choice the data do not make.

5. Sampling

The sampler (`gibbs.py`, a C kernel run from Python) is an exact Gibbs sampler: it updates one rupture at a time, drawing its rate from its exact conditional distribution by a slice sampler in log rate, with an exact step that switches the rupture on or off. Under the band each update is restricted to the rates that keep every row the rupture touches inside the band. Swap moves

exchange the rates of near-duplicate ruptures (1,165,613 pairs; 0.7% of swaps accepted), and the fraction on in each bin is redrawn after every sweep. The chains start from the best fit moved inside the band by a penalized least-squares fit (about 4 minutes; χ^2 93.5, no rows outside). Each of the three replicates ran 16 chains for 120,000 sweeps on 4 threads in about 3.0 hours, keeping 3,438 draws per chain after burn-in: 165,024 recorded draws in all. No recorded draw left the band. The runs keep running summaries rather than every draw (the full draws would take about 68 GB per replicate), plus 1,000 whole draws per replicate for the representative models and the figures.

6. Convergence

`check_convergence.py` judges each replicate against the targets of its preset, and all three **pass**. For χ^2 and the total rate the effective sample sizes are about 40,000 and 48,000 per replicate, and for the rupture rates the 5th percentile is 46,700. About 1,541 ruptures are on in each draw ($\hat{R}$ 1.0002–1.0004), and every bin's fraction on has $\hat{R}$ at most 1.0035. With 204,538 rates, even perfectly independent draws give a worst $\hat{R}$ above 1, so $\hat{R}$ is judged against that noise baseline (1.0004 here). A small minority of ruptures mixes slowly, mostly ruptures that switch on and off rarely: 15 to 27 ruptures per replicate (at most 0.013%) with $\hat{R}$ above 1.01 (worst 1.06), carrying 1.3–2.4% of the total rate, still with at least 499 effective draws each. The replicates agree rupture by rupture as independent runs should: the standardized differences of their mean log rates have a standard deviation of 0.986 (1 expected), with 0.22–0.23% beyond 3 (0.27% expected).

replicate	χ^2 (mean)	effective draws: χ^2 / total rate	worst $\hat{R}$ (baseline)	slow-mixing ruptures (share of rate)	verdict
main_rep0	438.5	40,870 / 47,430	1.053 (1.0004)	22 (1.4%)	PASS
main_rep1	438.5	40,470 / 48,800	1.055 (1.0004)	15 (1.3%)	PASS
main_rep2	438.5	39,270 / 48,220	1.060 (1.0004)	27 (2.4%)	PASS

7. Fit to the data

The posterior's χ^2 is 438.5 ± 20.3, below the fit check's 590 ± 34 (−4.4 standard deviations). The fit check assumes the Gaussian likelihood alone; the band is stricter than the data wherever a target is weak (a section whose target is less than about one uncertainty above zero would otherwise accept a prediction of zero), so under the band χ^2 sits below it, and the fit check no longer tests the uncertainties. The row-block fit (`block_fit.py`) splits χ^2 by data type. The slip rates are fitted more closely than their uncertainties imply, while the MFD bins and the paleo rows are fitted about twice as poorly as theirs: the posterior trades some of the bins' and paleo rows' fit for the slip rates. In the table, the "mean model" is the misfit of the posterior-mean prediction and the "spread" is the posterior variance of the predictions; the two add up to χ^2.

row block	rows	measurements	χ^2 over the draws	per measurement	z	mean model	spread
slip rates	2,312	507	271.5 ± 15.4	0.54	−7.4	147.8	123.7
MFD bins	18	18	44.7 ± 10.2	2.49	+4.5	28.2	16.5
paleo rates	65	65	122.2 ± 11.8	1.88	+5.0	101.0	21.3
total	2,395	590	438.5 ± 20.3	0.74	−4.4	276.9	161.5

8. Posterior tables, rate vectors and representative models

`make_tables.py` pools the three replicates into per-rupture, per-subsection and per-bin tables and into rate vectors (`rate_vectors.csv`): for each of the 211,042 ruptures its posterior mean, median, mode and 2.3%, 15.9%, 84.1% and 97.7% points. Most ruptures are off in most draws, so their medians and modes are zero; sums such as fault or bin rates must be built from the means. The posterior-mean total rate is 0.0419 per year, 12.6% above the bin targets and almost exactly

NZSHM22's 0.04187; the bin totals sit a median 14.5% above their targets (1.5% to 26.6%), against the best fit's 8.2%. About half of each subsection's posterior rate is carried by ruptures whose rates are individually below what the data can detect. `select_draws.py` then picks 100 weighted representative models from the 3,000 kept draws, clustering them by their rates per subsection and magnitude bin, for use in hazard calculations. With 100, the weighted mean total rate is within 0.7% of the exact mean and the worst bin within 2.8%, but only 78% of subsection means are within 10% (the tolerance asks for 90%), so the posterior-mean rate vector remains the exact input for mean hazard.

9. The posterior's fit, data type by data type

The next three figures repeat Figures 2–4 for the posterior. Each colored dot is the posterior mean, and each faint x one of the 3,000 kept draws, so where the draws gather shows darkest; open black squares are NZSHM22. **Rates by magnitude** (Figure 5): the posterior spreads each bin's rate over many ruptures, most of them with mean rates of 10^{-9} to 10^{-6} per year, where the best fit put a few ruptures high and the rest at zero. Every bin total lies above its target, by 1.5–4.6% below M7.1 and 11–27% from M7.1 upward (12.6% in all), a pattern NZSHM22's bin totals share; each draw's bin totals scatter about ±25% around the mean. **Slip rates** (Figure 6): 94% of the 507 sections' posterior means lie within one measurement uncertainty of their targets and 99% within two (NZSHM22: 100%), with a median ratio of prediction to target of 0.80. The x's show how much each section's prediction varies from model to model, often more than its measurement uncertainty for weakly constrained sections. **Paleo rates** (Figure 7): the posterior underpredicts them, with a median ratio of 0.61; 48% of rows lie within one uncertainty and 89% within two (mean-model χ^2 101.0, against NZSHM22's 72.6). This shortfall, which appears under every prior variant tried at 211,000 ruptures, comes from the tension between the paleo rows and the MFD bins and from the posterior spreading each bin's rate across its ruptures' magnitudes.

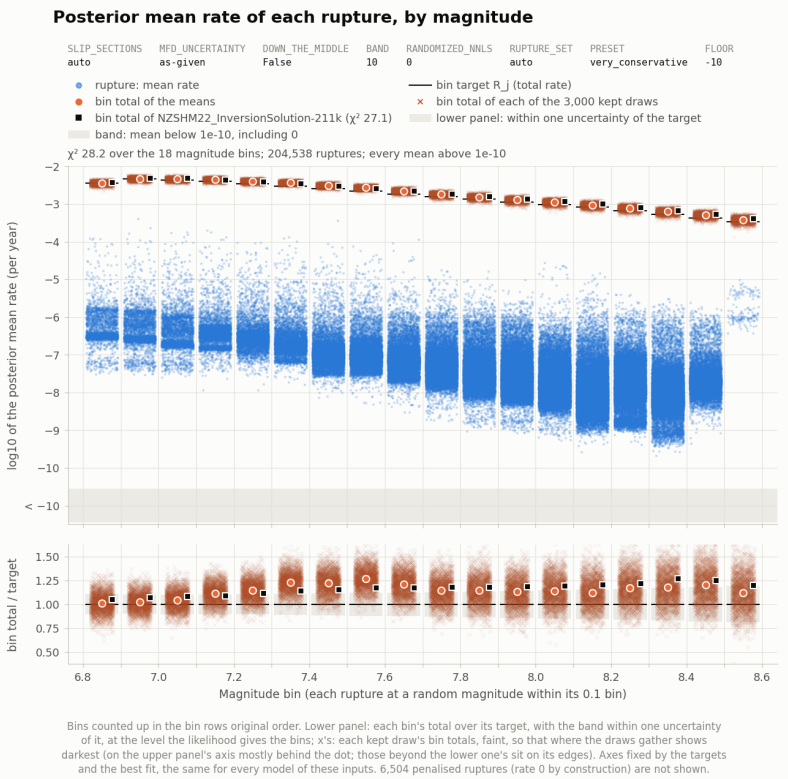


Figure 5. Posterior: each rupture's posterior mean rate against its magnitude bin, each bin's target (black line) and total of the means (orange dot), and NZSHM22's bin totals (black squares). Orange x's: each kept draw's bin totals. Lower panel: each bin's total over its target. `spikeslab/fig_rate_magnitude.png`

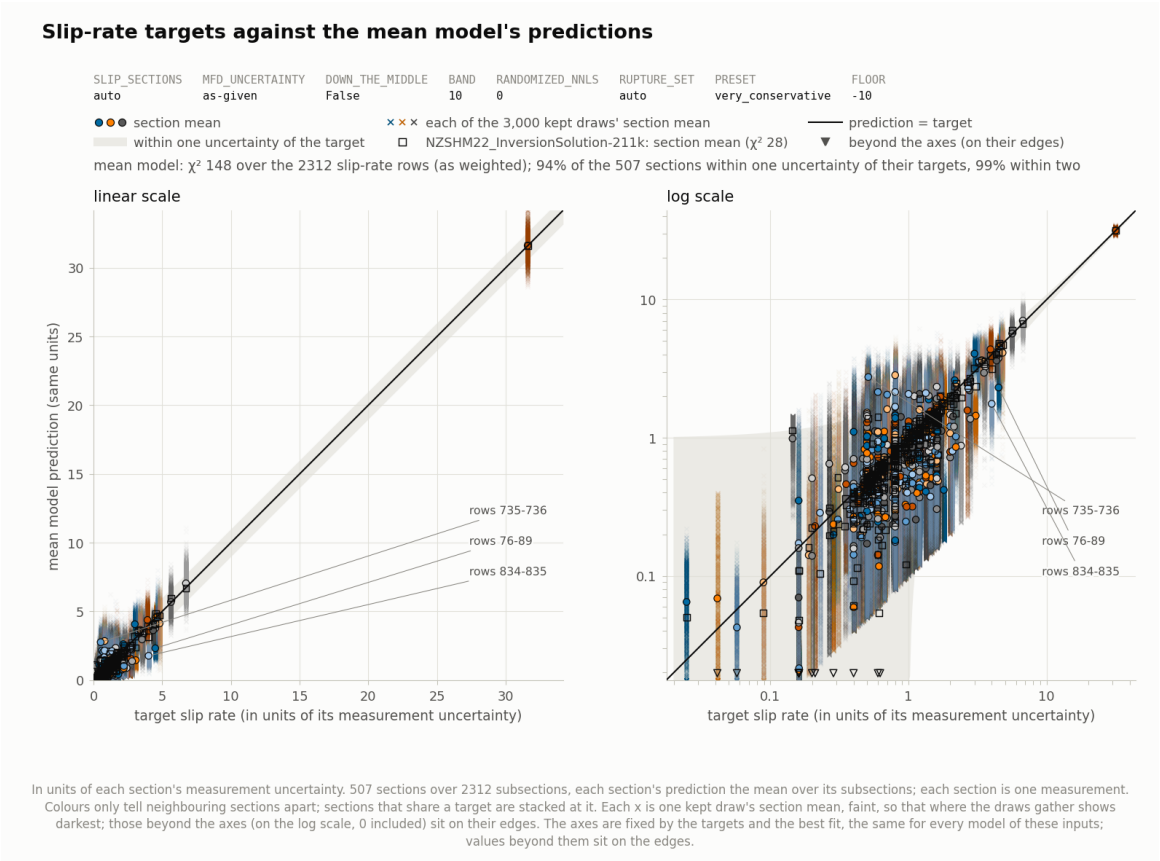


Figure 6. Posterior: each fault section’s slip-rate target against its posterior-mean section mean (dots) and each kept draw’s (x’s), in units of its measurement uncertainty; open squares are NZSHM22. The three sections furthest from their targets are labelled with their original rows. `spikeslab/fig_slip_rates.png`

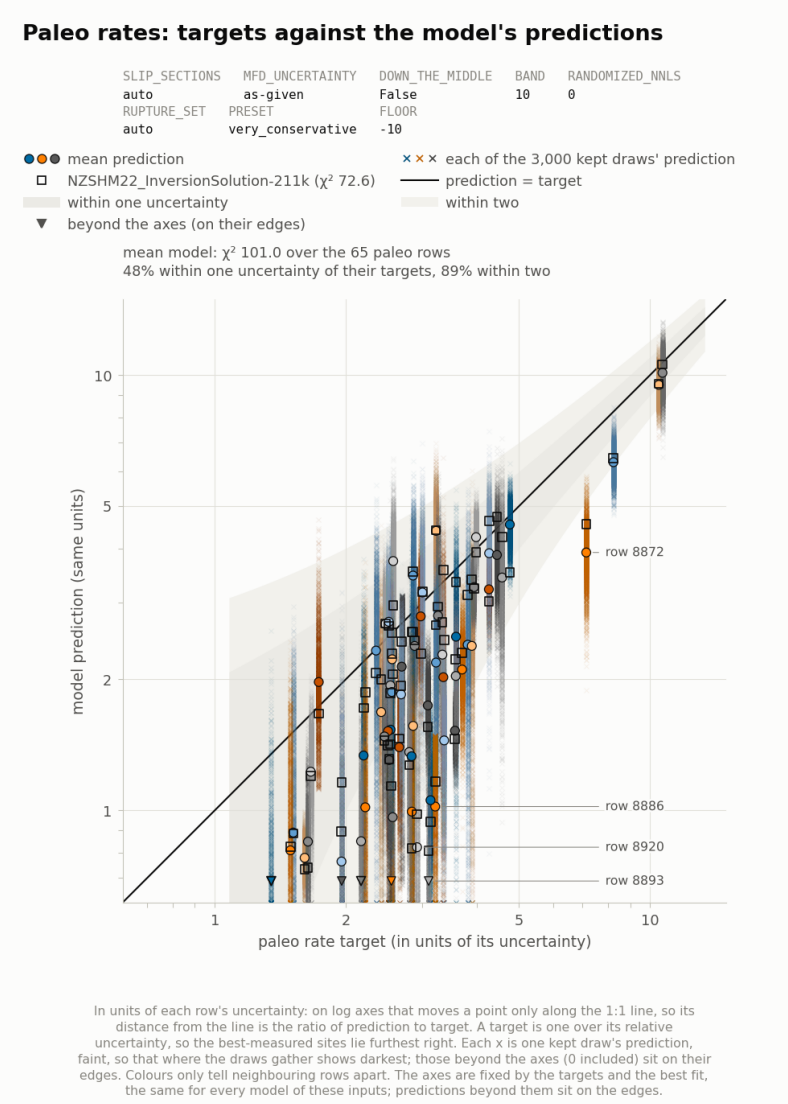


Figure 7. Posterior: each paleo row's target against its posterior-mean prediction (dots) and each kept draw's (x's); open squares are NZSHM22. The four rows furthest from their targets are labelled.
`spikeslab/fig_paleo_rates.png`

10. Faults one by one

`plot_slip_sections.py` looks at the 33 named faults of the rupture set's `named_faults.json`, ordered from the most isolated (Greendale, which shares no rupture with any other section) to the most connected (Hope, which shares ruptures with 125 other sections). **Figure 8** shows each fault's mean slip-rate prediction against its target. Under the band every named fault has an event in every draw (the middle panel is 0% throughout); most posterior means lie within one uncertainty of their targets, the isolated faults with the widest spread between draws. The largest shortfall is Akatore (0.02 against a target of 0.16, in units of its uncertainty), which NZSHM22 predicts equally low. **Figures 9 and 10** count each fault's ruptures by rate. The posterior means spread rate over hundreds to tens of thousands of ruptures at low rates, whereas any single draw switches on only a few ruptures per rate bin, at higher rates, much as NZSHM22 (black outline) does: a single model compares with a single draw, not with the mean. **Figures 11 and 12** sum each fault's rupture rates by magnitude bin, its rate of events of each size. The circles are the posterior means, the x's the kept draws' sums, and the strip marked 0 shows by its opacity how often a draw has no event from that fault in the bin. NZSHM22's magnitude distribution mostly lies within the spread of the draws, but where it concentrates a fault's rate in a few bins the posterior spreads it over all the magnitudes the fault's ruptures allow.

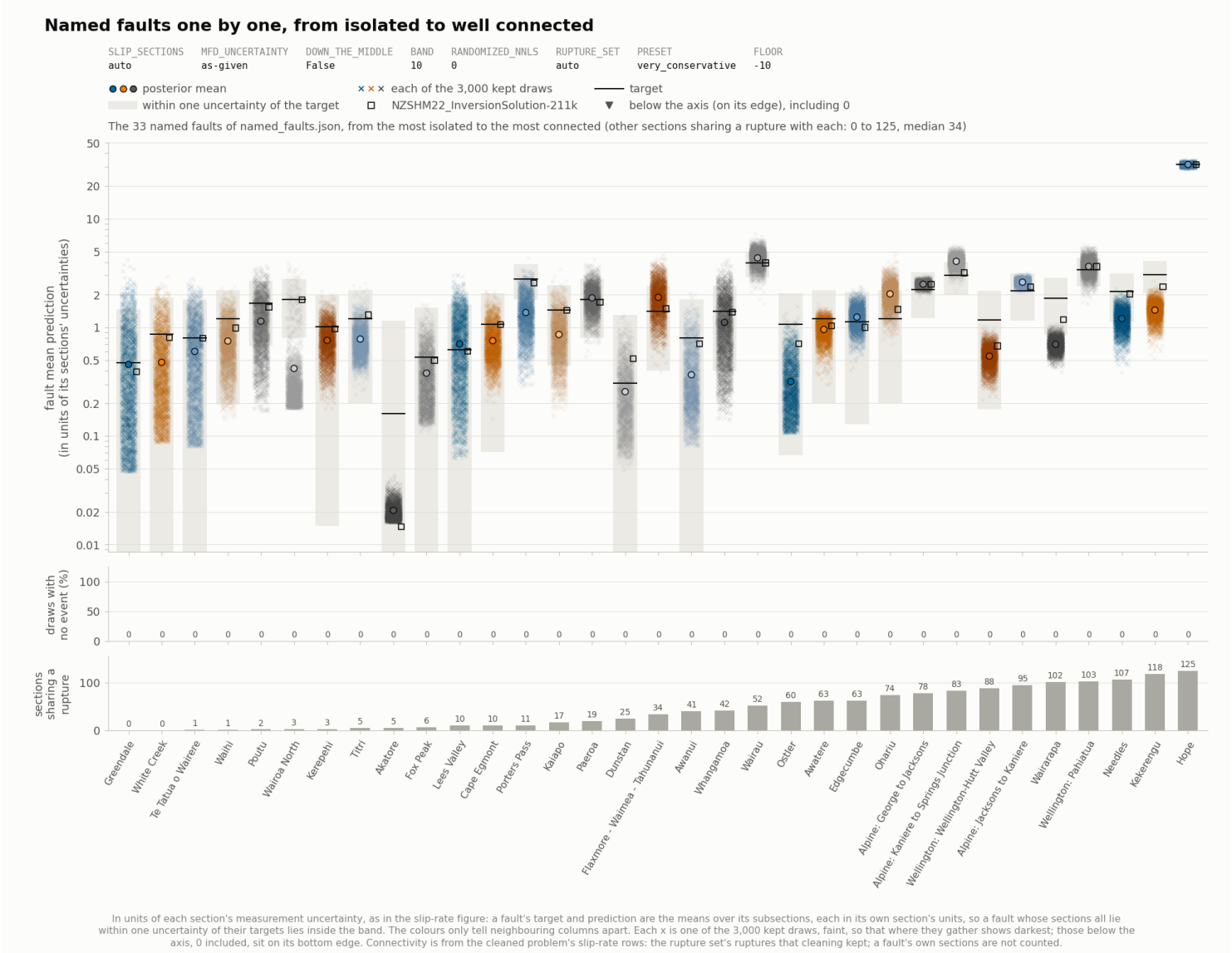


Figure 8. The 33 named faults, from the most isolated to the most connected: each fault's mean slip-rate prediction over its subsections (posterior mean: dot; kept draws: x's; NZSHM22: open square) against its target (black line; grey: within one uncertainty), in units of its sections' uncertainties. Below: the share of draws in which no rupture crossing the fault is on, and the number of other sections sharing a rupture with it. `spikeslab/fig_slip_sections.png`

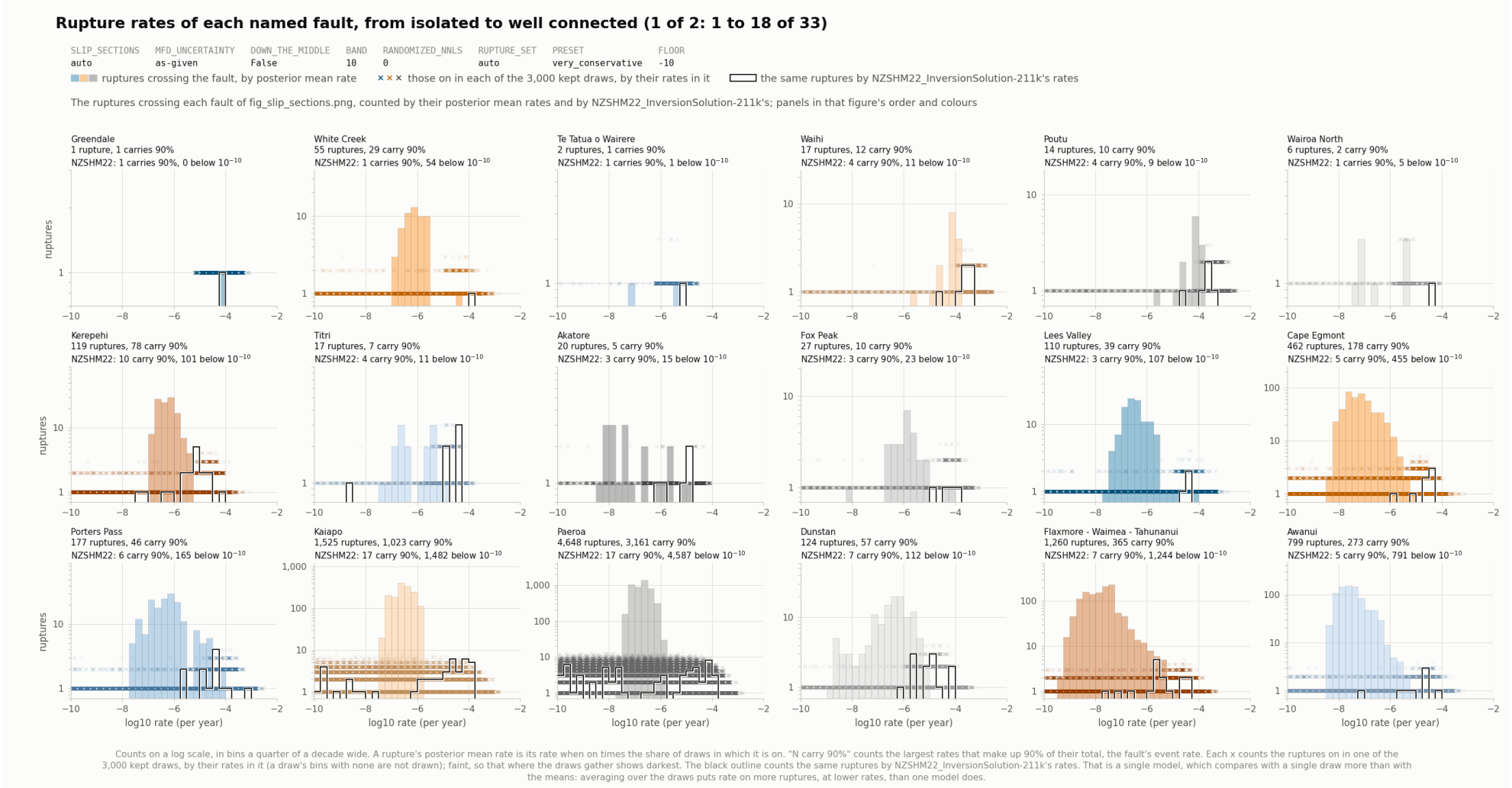


Figure 9. Rupture rates of the first 18 named faults: the ruptures crossing each fault counted by their posterior mean rates (light bars), each kept draw's counts of the ruptures on in it (x's), and NZSHM22's rates of the same ruptures (black outline). `spikeslab/fig_slip_sections_ruptures_1.png`

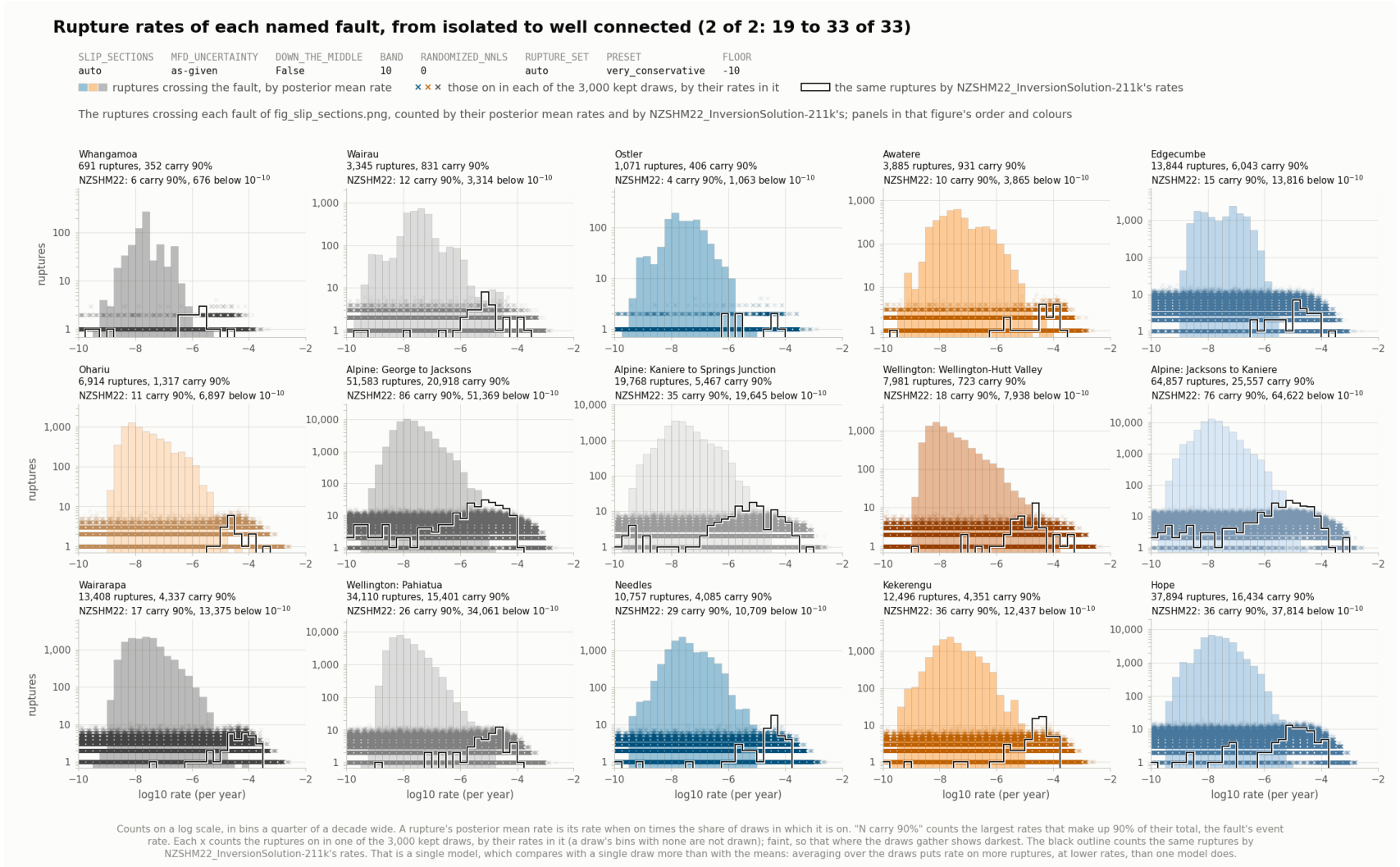


Figure 10. As Figure 9, for the last 15 named faults. `spikeslab/fig_slip_sections_ruptures_2.png`

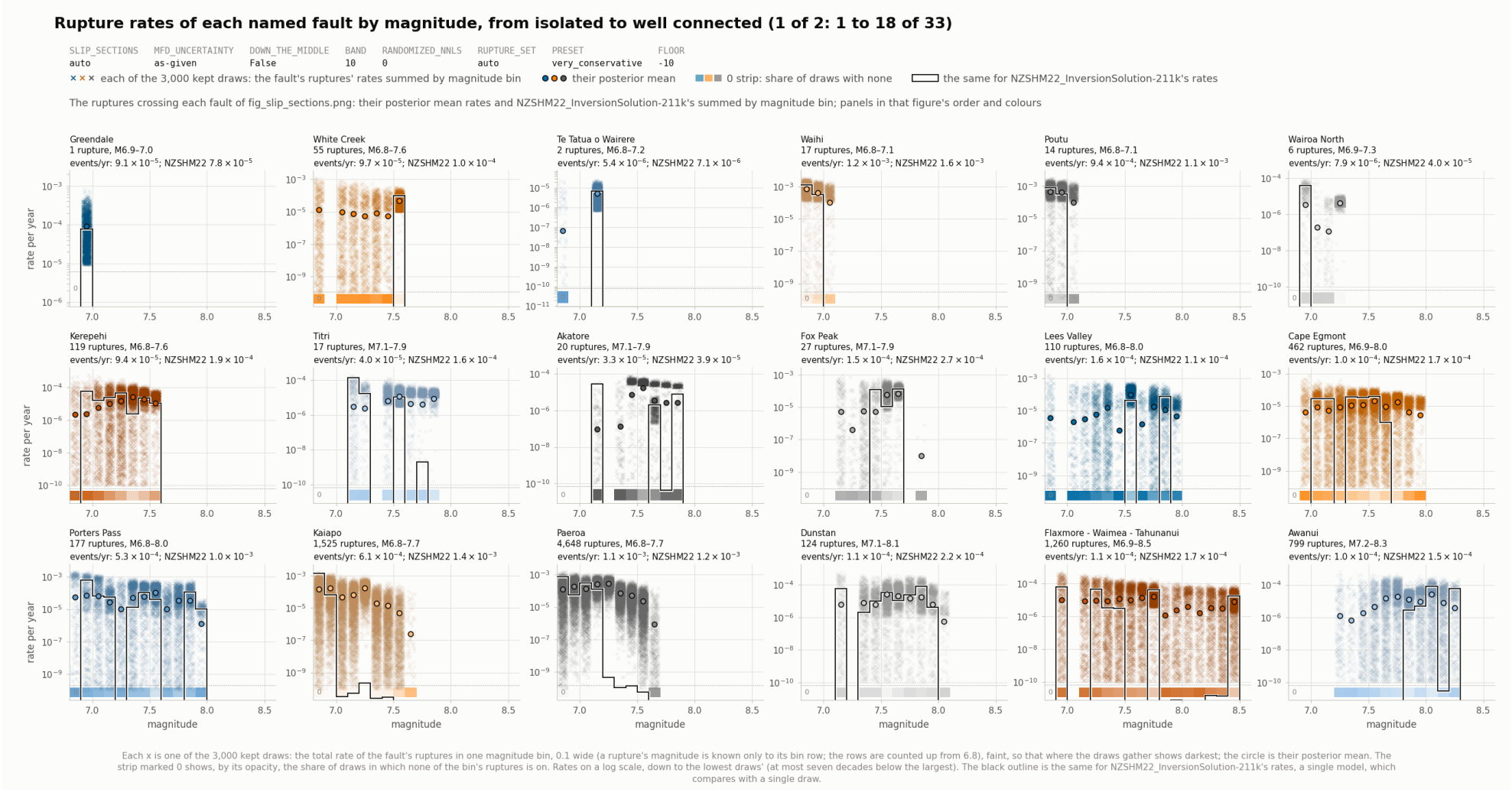


Figure 11. Rupture rates of the first 18 named faults by magnitude: each kept draw's rates of the fault's ruptures summed by magnitude bin (x's), their posterior mean (circles), the share of draws with no event in each bin (opacity of the strip marked 0), and NZSHM22's sums (black outline).

`spikeslab/fig_slip_sections_ruptures_by_magnitude_1.png`

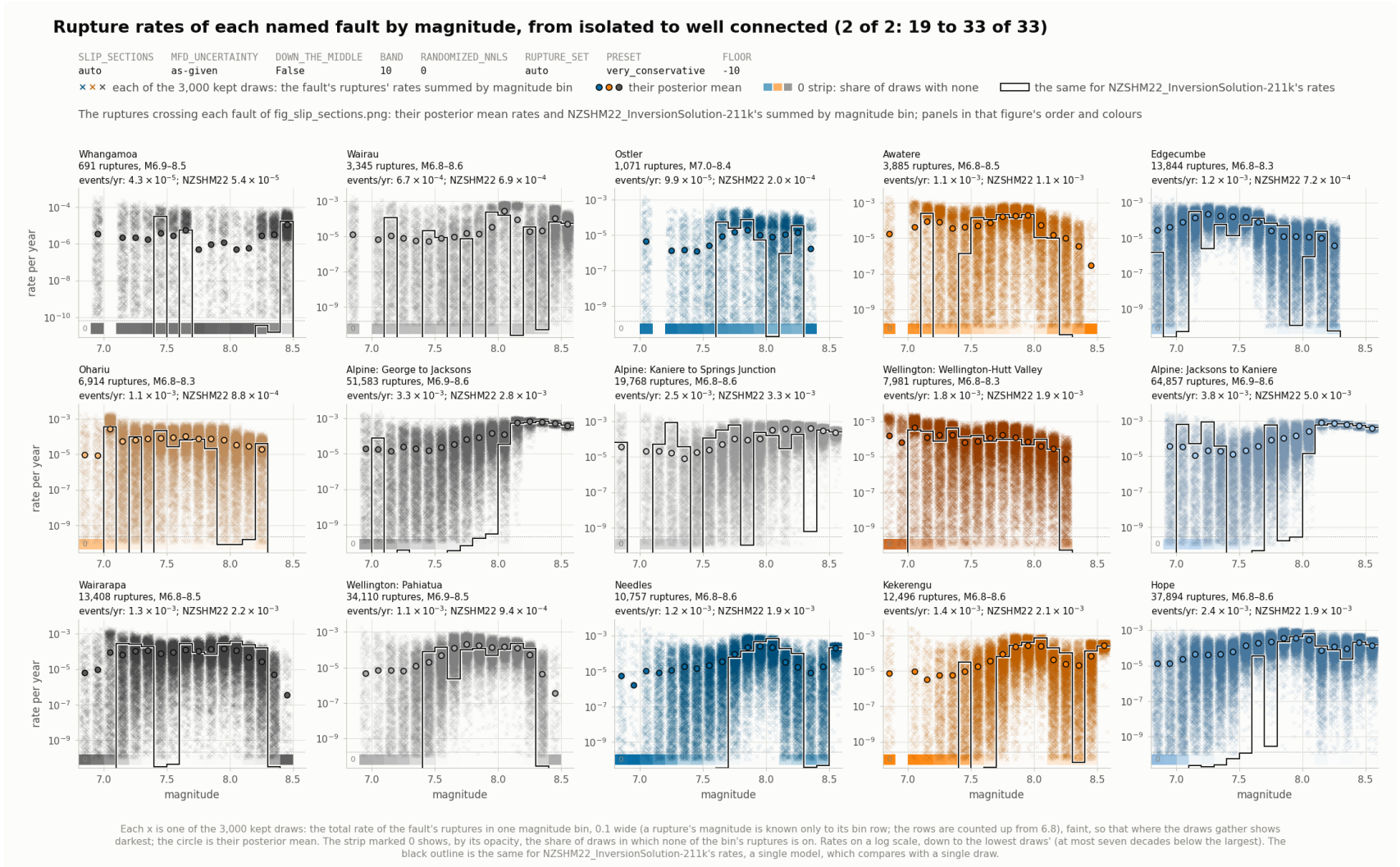


Figure 12. As Figure 11, for the last 15 named faults. `spikeslab/fig_slip_sections_ruptures_by_magnitude_2.png`

11. Output for OpenSHA

`make_solution.py` writes the posterior-mean rates as an OpenSHA fault-system solution's rates file (`spikeslab/rates.csv`, one row per rupture of the rupture set, removed ruptures at zero), and the best fit's as `nnls/rates.csv`. Before writing, it checks that the rates and the rupture set describe the same ruptures: all 204,538 kept ruptures cross exactly the fault sections the rupture set lists, and each one's magnitude lies in its bin. With `--zip` it also writes a complete archive that OpenSHA loads. Any representative model can be written the same way.

12. Points to keep in mind

• **The fit check under the band.** χ^2 is 4.4 standard deviations below the fit check, because the band is stricter than the data where targets are weak; a fit check under the band would need its expected χ^2 from simulated data. The factor 10 is a modelling choice, and a run at another factor (3, say) would show how much the results depend on it.
• **Paleo rates.** The posterior mean underpredicts them (median ratio 0.61), as at every prior setting tried at this size.
• **Slow mixing.** 15–27 ruptures per replicate, carrying 1.3–2.4% of the total rate, mix slowly; their individual rates are less precise than the rest, though sums over many ruptures are not affected.
• **Starts.** All 48 chains started from the same point (the best fit moved inside the band); runs from dispersed starts remain untested.
• **Representative models.** The 100 representatives miss some of their tolerances on subsection rates; the posterior-mean rate vector is the exact input for mean hazard.
• **What the data determine.** Sums of rates (fault, subsection and bin rates) and the probability that a rupture exceeds about 10^{-5} per year are well determined; individual rates below that are largely set by the prior.